Radiation uses & nuclear energy

Compare exposure

1 Exposure routes

Irradiation: source exposes object; remove/shield source → exposure ends.

Contamination: unwanted radioactive material remains and decays. Alpha is especially harmful internally; gamma can irradiate from outside.

2 Dose & precautions

Reduce time, increase distance, use suitable shielding and prevent inhaling/swallowing material.

Background: rocks/radon, cosmic rays, living things; fallout adds human-made sources. Location/work affect dose. **1 Sv = 1,000 mSv.**

Choose a source

3 Useful emissions

Gamma tracer escapes body; choose half-life long enough to measure, short enough to limit dose.

Target radiation to destroy unwanted tissue while limiting healthy-tissue damage. Beta can monitor thin-sheet thickness; gamma sterilises equipment.

4 Judge risks

Same nuclei count: shorter half-life → greater initial activity, faster decline. Long-lived contamination persists.

Consider type, dose, position and time with benefit and alternatives. Peer review checks research claims.

Release nuclear energy

5 Fission

Large unstable nucleus usually absorbs a neutron and splits into two smaller nuclei.

Releases 2–3 neutrons, gamma and energy; products have kinetic energy. Neutrons can trigger more fissions; reactor controls the chain.

6 Fusion

Two light nuclei join into a heavier nucleus; some mass becomes radiation energy.

High temperatures help overcome repulsion; powers stars. Not the same as melting. Both fission and fusion change nuclei.